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Original article

# Quantification of adipose tissues by Dual-Energy X-Ray Absorptiometry and Computed Tomography in colorectal cancer patients

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SUMMARY

*Background & aims:* Excess adipose tissue may affect colorectal cancer (CRC) patients' disease progression and treatment. In contrast to the commonly used anthropometric measurements, Dual-Energy X-Ray Absorptiometry (DXA) and Computed Tomography (CT) can differentiate adipose tissues. However, these modalities are rarely used in the clinic despite providing high-quality estimates. This study aimed to compare DXA's measurement of abdominal visceral adipose tissue (VAT) and fat mass (FM) against a corresponding volume by CT in a CRC population. Secondly, we aimed to identify the best single lumbar CT slice for abdominal VAT. Lastly, we investigated the associations between anthropometric measurements and VAT estimated by DXA and CT.

*Methods:* Non-metastatic CRC patients between 50-80 years from the ongoing randomized controlled trial CRC-NORDIET were included in this cross-sectional study. Corresponding abdominal volumes were acquired by Lunar iDXA and from clinically acquired CT examinations. Also, single CT slices at L2-, L3-and L4-level were obtained. Agreement between the methods was investigated using univariate linear regression and Bland–Altman plots.

*Results:* Sixty-six CRC patients were included. Abdominal volumetric VAT and FM measured by DXA explained up to 91% and 96% of the variance in VAT and FM by CT, respectively. Bland–Altman plots demonstrated an overestimation of VAT by DXA compared to CT (mean difference of 76 $cm^3$) concurrent with an underestimation of FM (mean difference of −319 $cm^3$). A higher overestimation of VAT ($p = 0.015$) and underestimation of FM ($p = 0.036$) were observed in obese relative to normal weight subjects. VAT in a single slice at L3-level showed the highest explained variance against CT volume ($R^2 = 0.97$), but a combination of three slices (L2, L3, L4) explained a significantly higher variance than L3 alone ($R^2 = 0.98$, $p < 0.006$). The anthropometric measurements explained between 31-65% of the variance of volumetric VAT measured by DXA and CT.

*Conclusions:* DXA and the combined use of three CT slices (L2-L4) are valid to predict abdominal volumetric VAT and FM in CRC patients when using volumetric CT as a reference method. Due to the poor

*Abbreviations:* CRC, colorectal cancer; DXA, Dual-Energy X-Ray Absorptiometry; CT, Computed Tomography; VAT, visceral adipose tissue; FM, fat mass; SAT, subcutaneous adipose tissue; IMAT, inter- and intramuscular adipose tissue; ISCD, International Society for Clinical Densitometry; PG-SGA, Patient-Generated Subjective Global Assessment.

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performance of anthropometric measurements we recommend exploring the added value of advanced body composition by DXA and CT integrated into CRC care.
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## 1. Introduction

Excess adipose tissue is an established risk factor for the development of colorectal cancer (CRC) [1,2], in addition to influence CRC prognosis [3,4] and treatment [5]. However, not all adipose tissues are alike as they have several structural and functional differences [6,7]. Visceral adipose tissue (VAT) is in general more metabolically active compared to subcutaneous adipose tissue (SAT) [8,9]. Also, inter- and intramuscular adipose tissue (IMAT) is associated with insulin resistance, loss of strength and mobility dysfunction [10,11]. After CRC surgery patients may experience a shift in body composition towards decreasing skeletal muscle mass and increasing amounts of adipose tissue [12]. Quantification of the individual's adipose tissues and muscle mass allows risk-stratification and personalized treatment [13,14].

Anthropometric measurements such as BMI are commonly used, but neither able to distinguish different adipose tissues from one another nor muscle [7]. Thus, appropriate clinical follow up of CRC patients calls for accurate tools that can monitor shifts in body composition, including different adipose tissues [12].

Advanced imaging methods like Computed Tomography (CT) and Dual-Energy X-Ray Absorptiometry (DXA) are examples of such tools. Abdominal CT is a part of standard oncological care, but is rarely used for body composition purposes in the clinic despite being considered a reference standard [15]. In research the assessment of adipose tissues by CT has typically been performed by a single lumbar slice and not by volumetric measures. This mainly due to the comprehensive post-processing in lack of automated quantifying software. Therefore, volumetric CT data consisting of multiple continuous single slices is, although considered a reference standard, unattainable in most settings. There is limited evidence on which lumbar landmark is superior for estimation of abdominal adipose tissues in a CRC population.

Although mainly used in research, DXA is a non-invasive, low-cost, and easily applicable method for studying body composition [15,16]. While CT provides three-dimensional imaging, DXA is two-dimensional [15]. Therefore, the DXA software uses geometrical assumptions to estimate VAT by subtracting the SAT layer from total fat mass (FM) in the abdominal area [17,18]. Due to the location of IMAT, deep to SAT, we hypothesize that DXA at least partly detect IMAT and misclassifies it as VAT. Following surgical resection, anatomical changes in CRC patients may challenge DXA's geometrical assumptions for distinguishing fat masses. Therefore, validation of DXA in a pure CRC population is necessary.

The aim of this study was to compare DXA's measurement of abdominal VAT and FM against a corresponding volume measured by CT in a CRC population. Secondly, we investigated the ability of single slices at the lumbar level (L2-L4) to predict abdominal volumetric VAT by CT ($VAT_{CT}$). Finally, we examined the association between anthropometric measurements (BMI, waist circumference, and waist/hip ratio) and abdominal $VAT_{DXA}$ and $VAT_{CT}$.

## 2. Materials and methods

### 2.1. Subjects and inclusion criteria

A sub-population from the ongoing randomized controlled trial, CRC-NORDIET [19], was enrolled in the current cross-sectional study. The CRC-NORDIET study included patients, 50–80 years of age, who were diagnosed with CRC stage I-III and treated surgically [19]. The CRC patients could both have finished treatment or received adjuvant treatment at the time of inclusion. The baseline in the CRC-NORDIET study takes place two to nine months post-operatively and includes a DXA scan. CT scans are not performed in the CRC-NORDIET study, but are clinically acquired as a part of the routine six months post-operative follow up at the hospital. Inclusion in the current study required 1) participation at baseline in the CRC-NORDIET study and 2) a maximum of 45 days between the DXA scan conducted at baseline and the routine CT scan performed six months postoperatively. The time limit of 45 days was imposed to avoid substantial changes in body composition between DXA and CT scan.

### 2.2. DXA

DXA scans were obtained using Lunar iDXA (GE Healthcare Lunar, Buckinghamshire, UK) and the software enCORE ™ version 16 including the application CoreScan. Subjects were scanned at baseline in CRC-NORDIET by trained personnel and the scan was performed in line with the protocol of the International Society for Clinical Densitometry (ISCD) [20]. The system was calibrated daily prior to scans and subjects were fasting and wearing light clothing. The region-of-interest (hereafter referred to as the abdominal volume) was individually defined as 20% of the distance from the iliac crest towards the lowest point of the mandible (gnathion) (Fig. 1). Abdominal volumes of VAT, SAT and FM were retrieved from the DXA scans.

### 2.3. CT

Clinically indicated CT scans were acquired at Oslo University Hospital, Ullevål and Akershus University Hospital as part of the routine follow up six months after CRC surgery. From these CT scans we extracted image volumes corresponding to the individually DXA-defined abdominal volume (Fig. 1). These image volumes consisted of 29–35 consecutive 3 mm axial CT slices. In addition to the entire volume, single axial mid-vertebral CT slices were extracted at the 2nd, 3rd and 4th lumbar vertebrae (L2, L3, and L4).

#### 2.3.1. Segmentation of CT slices

Segmentation of VAT, IMAT, SAT and SM in CT slices was performed (Fig. 1) with the sliceOmatic software package (v 5.0 rev. 7b, Tomovision, Magog, QC, Canada) in accordance with the Alberta protocol. Abdominal volumetric $FM_{CT}$ was defined as the sum of $VAT_{CT}$, $IMAT_{CT}$ and $SAT_{CT}$. An in-house software based on

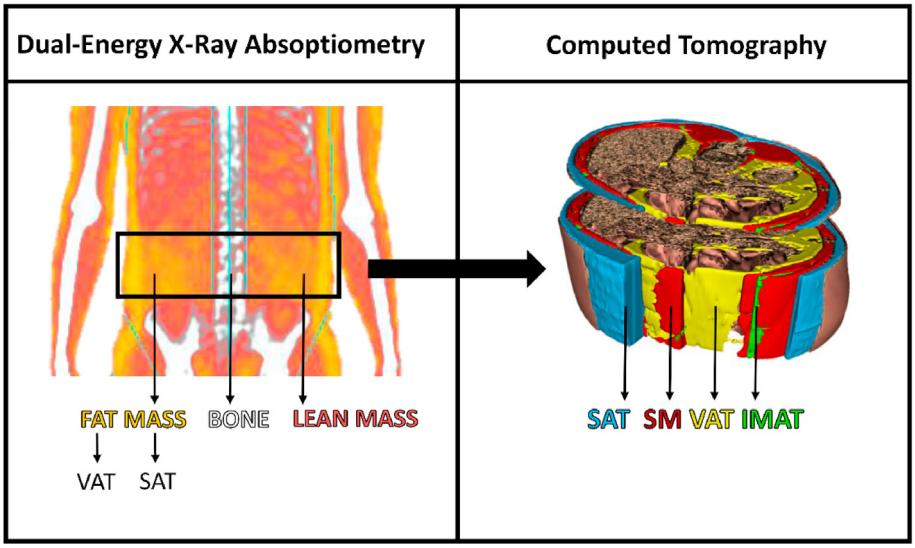


**Fig. 1.** An illustration of the various body compartments in the corresponding abdominal volume obtained by DXA (illustrated as the area inside the black rectangle) and CT, respectively. Abbreviations: DXA, Dual-Energy X-ray Absorptiometry; CT, Computed Tomography; VAT, visceral adipose tissue; SAT, subcutaneous adipose tissue; IMAT, inter- and intramuscular adipose tissue; SM, skeletal muscle mass.

convolutional neural networks, BodySegAI, was used for automatic and more efficient segmentation process. Quality assurance with semi-manual corrections was conducted after segmentation by BodySegAI, as this software was under validation. All CT data were segmented by a single operator (DHA) under the supervision of an experienced radiologist (PML).

### 2.4. Anthropometric measurements

BMI, waist circumference and waist/hip ratio were performed in accordance with standardized protocols [19]. Data on self-reported weight change were obtained from the Patient-Generated Subjective Global Assessment (PG-SGA), a nutrition assessment tool validated for use in cancer patients [21,22]. Weight change data were only used for subjects that performed a CT scan prior to the DXA scan, as the PG-SGA is retrospective and was completed the same day as the DXA scan. Subjects with ≤7 days between the DXA- and CT scan were classified as weight stable, which was defined as alterations in weight less than ±1 kg.

### 2.5. Statistical analyses and sample size calculation

Data were analyzed using SPSS (IBM SPSS Statistics 25). Normality was checked by visual inspection of histograms and Q–Q plots and tested using the Shapiro–Wilk normality test. Characteristics of the study population are presented by median and percentiles ($25^{th}$ and $75^{th}$) and count (n) and percentage. Scatter plots with coefficients of determination ($R^2$) were constructed to examine the degree of covariance between fat masses estimated by DXA, CT and anthropometric measurements. To quantify potential differences between volumetric fat masses by DXA and CT, as well as visualize potential patterns in data, Bland–Altman plots with predetermined limits of agreement were created (mean difference±1.96xSD) [23]. Comparisons of $VAT_{DXA}$ and $VAT_{CT}$ were performed with and without the inclusion of $IMAT_{CT}$ in $VAT_{CT}$ and tested using the Paired Samples T-test. The Mann–Whitney U-test with Bonferroni correction was used to examine differences between DXA and CT measurements across median VAT or BMI within the boxplots. The whiskers represents the minimum and maximum values (Q1/Q3 ± (1.5x interquartile range)). Regression equations and Pearson's correlation coefficients were provided for the associations. When predicting the difference between methods, the mean difference was expressed on the y-axis and mean VAT or BMI on the x-axis. Pairwise comparisons by Meng, Rosenthal, and Rubin were performed to examine the potential differences between correlations for the single slices [24]. Bonferroni corrections were applied. The sample size calculation was based on correlation coefficients from previous validation studies of $VAT_{DXA}$ [2,18,25]. One-sided z-test for Fisher's z-transformation of r was used to calculate the sample size required for detecting a Pearson's correlation coefficient of r = 0.9 or higher, with a power of 0.8 and a significance level of alpha = 0.05. Assuming the true correlation was >0.95, a sample size of 51 subjects was sufficient.

### 2.6. Ethical statement

The CRC-NORDIET study was approved by the Regional Committees for Medical and Health Research Ethics (REC Protocol Approval 2011/836) and the data protection officials at Oslo University Hospital and Akershus University Hospital. The study is carried out in accordance with the Helsinki Declaration and informed consents were obtained from the participants. The CRC-NORDIET study is registered on the National Institutes of Health Clinical Trials (www.ClinicalTrials.gov; Identifier: NCT01570010).

## 3. Results

### 3.1. Patient characteristics

Hundred-and-thirty-five subjects were available at baseline in the CRC-NORDIET study. Fifty-two were excluded as they exceeded the 45 day limit between baseline DXA and postoperative CT scan. Another 17 subjects were excluded due to lack of CT-data in electronic patient records leaving a total of 66. Characteristics of the study population are shown in Table 1. The median age was 64 years, and the median BMI was 26.5 kg/$m^2$ for men and women combined. The median number of days between the CT and DXA scan was 17 days. Weight change data were available for 70% of the subjects, and most reported weight stability.

**Table 1**
The characteristics of study population.

| | Total n = 66 | | Male n = 31 | | Female n = 35 | |
|---|---|---|---|---|---|---|
| | Median | p25, p75 | Median | p25, p75 | Median | p25, p75 |
| Age (y) | 64.0 | (61,68) | 64.0 | (61,68) | 64 | (60,71) |
| BMI (kg/m$^2$) | 26.5 | (23,29) | 27.0 | (25,30) | 26.3 | (23,30) |
| Weight (kg) | 77.9 | (69,93) | 85.5 | (77,99) | 72.0 | (66,81) |
| Waist circumference (cm) | 94.1 | (85,103) | 98.5 | (94,110) | 86.5 | (81,95) |
| Waist/hip ratio | 0.9 | (0.9,1.0) | 1.0 | (1.0,1.0) | 0.9 | (0.8,0.9) |
| $VAT_{CT}$ (cm$^3$) [1] | 994 | (560,1786) | 1754 | (973,2454) | 662 | (423,1227) |
| $VAT_{DXA}$ (cm$^3$) [1] | 1230 | (707,1996) | 1706 | (1082,2610) | 769 | (623,1310) |
| $FM_{CT}$ (cm$^3$) [1] | 2954 | (2199,4255) | 3361 | (2669,4403) | 2596 | (1867,3501) |
| $FM_{DXA}$ (cm$^3$) [1] | 2511 | (2005,3524) | 2956 | (2322,3766) | 2378 | (1782,3336) |
| $IMAT_{CT}$ (cm$^3$) [1] | 102 | (66,153) | 102 | (74,164) | 102 | (62,151) |
| Days between DXA and CT scan | 16.5 | (6,34) | 14.0 | (7,34) | 19.0 | (6,37) |
| Days between DXA scan and cancer operation | 189 | (154,221) | 188 | (152,211) | 193 | (176,226) |
| | n (%) | | n (%) | | n (%) | |
| Self-reported weight change [2] | | | | | | |
| Reduced | 0 (0) | | 0 (0) | | 0 (0) | |
| Stabile | 38 (58) | | 15 (48) | | 23 (66) | |
| Increased | 8 (12) | | 3 (10] | | 5 (14] | |
| Unknown | 20 (30) | | 13 (42) | | 7 (20) | |
| Tumour localization | | | | | | |
| C18 Colon | 41 (62) | | 17 (55) | | 24 (69) | |
| C19 Rectosigmoid | 5 (8) | | 4 (13) | | 1 (3) | |
| C20 Recti | 20 (30) | | 10 (32) | | 10 (29) | |
| TNM stage | | | | | | |
| I | 14 (21) | | 8 (26) | | 6 (17) | |
| II | 23 (35) | | 13 (42) | | 10 (29) | |
| III | 29 (44) | | 10 (32) | | 19 (54) | |
| Received adjuvant chemotherapy | 5 (8) | | 3 (10) | | 2 (6) | |

Median and percentiles (25th, 75th), count (n) and percent of total (%) are presented. [1] Results given for the abdominal volume. [2] Weight change defined as ±1 kg one month before the DXA scan, information obtained from the PG-SGA. Abbreviations: VAT, visceral adipose tissue; CT Computed Tomography; DXA, Dual-Energy X-ray Absorptiometry; FM, fat mass; IMAT, inter- and intramuscular adipose tissue; TNM stage, tumour, node and metastasis stage.

### 3.2. Assessment of VAT and FM by DXA and CT

Scatter plots with coefficients of determination ($R^2$) and Bland–Altman plots of $VAT_{DXA}$ against $VAT_{CT}$ (excluding and including IMAT) are presented in Fig. 2. The coefficient of determination between abdominal volumetric $VAT_{DXA}$ and $VAT_{CT}$ was $R^2 = 0.89$ excluding IMAT, and $R^2 = 0.91$ IMAT included. The Bland–Altman plots demonstrated a mean overestimation of $VAT_{DXA}$ compared to $VAT_{CT}$ of 201 cm$^3$. The mean difference between methods was significantly reduced when including IMAT to $VAT_{CT}$ in the comparison with $VAT_{DXA}$ (from 201 cm$^3$ to 76 cm$^3$, $p < 0.001$). The Bland–Altman plots illustrated increased variation between the methods for VAT above 1000 cm$^3$.

Scatter plot and Bland–Altman plot for abdominal volumetric FM between DXA and CT is presented in Fig. 3. The coefficient of determination between methods was $R^2 = 0.96$. The Bland–Altman plot shows a mean underestimation by DXA of −319 cm$^3$ for abdominal underestimation with the increasing size of the FM compartment.

To examine whether DXA's accuracy for VAT and FM measurements was related to the level of body fatness, differences between DXA and CT were investigated across BMI and VAT categories (Fig. 4). Fig. 4A shows the difference between methods across VAT categories. There were no significant differences between groups (all p-values>0.05). Fig. 4B shows the difference between methods across BMI categories. DXA demonstrated a significantly higher overestimation of VAT in obese versus normal weight subjects (median and 25th,75th percentile: 397.0 cm$^3$ (120.2, 741.5) vs. 77.3 cm$^3$ (−116.7, 265.1), $p = 0.015$). Concurrently, an increased underestimation of FM was seen in obese versus normal weight subjects (median and 25$^{th}$, 75$^{th}$ percentile: −426.8 cm$^3$ (−760.7, −173.6) vs. −128.2 (−300.1, −42.1), $p = 0.036$).

Linear regression analysis showed that with every 100 cm$^3$ increase in mean VAT (IMAT not included) the difference between DXA and CT increased with 16 cm$^3$ ($y = 17.39 + 0.16x$, $r = 0.44$, significant regression equation p-value<0.001). Regression equations for differences between methods for VAT (IMAT not included) and FM across BMI showed that for every unit increase in BMI, mean difference between DXA and CT increased with 31.8 cm$^3$ for VAT ($y = -663.5 + 31.8x$, $r = 0.45$, significant regression equation p-value<0.001), and −18.4 cm$^3$ for FM ($y = 180.2-18.4x$, $r = 0.30$, significant regression equation $p = 0.014$).

### 3.3. CT single slice predictions

Associations between measurements of VAT single slices against abdominal volumes by CT and DXA are presented in Table 2. Overall, the single slice at L3-level demonstrated the highest explained variance among the single slices against the corresponding abdominal volumetric alternative by CT and DXA ($R^2 = 0.87–0.97$). However, a combination of three slices at L2, L3-and L4-level demonstrated statistically significant higher explained variance ($R^2 = 0.90–0.98$) than from a single L3 slice ($p = 0.006$-<0.001 for all comparisons).

### 3.4. Anthropometric measurements

The relationships between BMI, waist circumference, and waist/hip ratio and abdominal volumes of $VAT_{CT}$ and $VAT_{DXA}$ are described in Fig. 5. BMI, waist circumference and waist/hip ratio explained 31, 64, and 52%, respectively of the variability in volumetric abdominal $VAT_{CT}$. Similarly, the anthropometric measurements described 39, 65 and 43%, respectively of the variability in abdominal volumetric $VAT_{DXA}$.

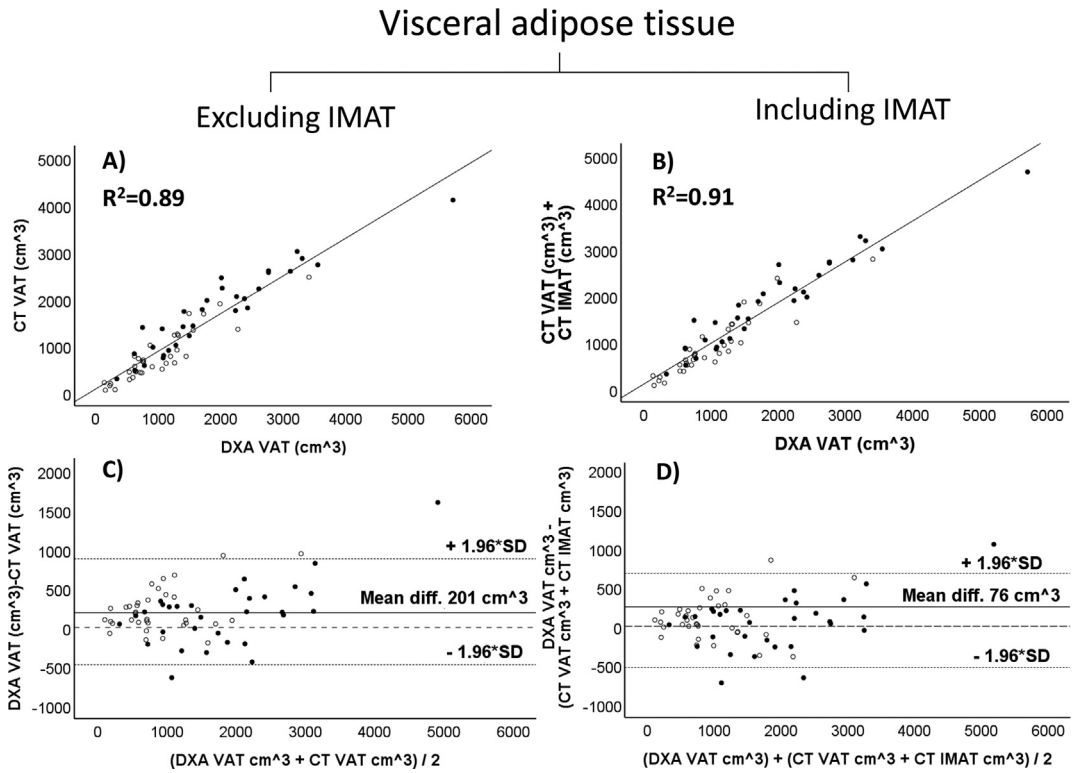


**Fig. 2.** Comparing DXA and CT in the assessment of abdominal volumetric VAT, excluding and including IMAT in scatter plots and Bland Altman plots. Men are annotated by black dots and females by white dots. In the Bland Altman plots the differences between methods are shown on the y-axis and mean VAT is on the x-axis. Abbreviations: VAT, visceral adipose tissue; IMAT, inter- and intramuscular adipose tissue, CT Computed Tomography; DXA, Dual-Energy X-ray Absorptiometry.

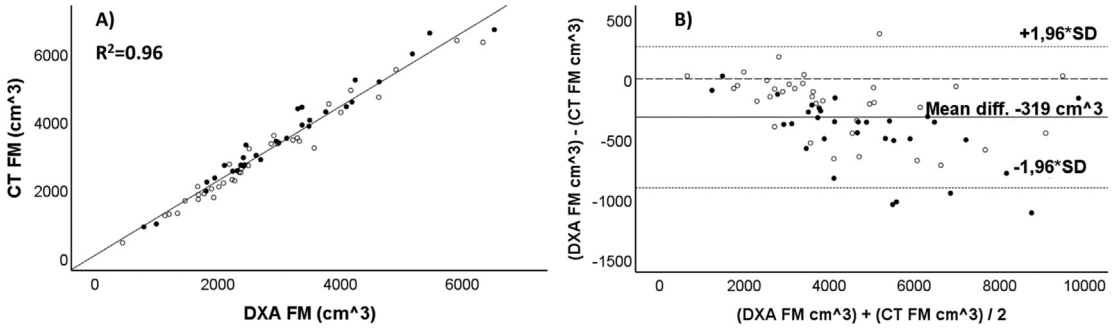


**Fig. 3.** Scatter plot with coefficient of determination ($R^2$) and a Bland–Altman plot comparing DXA and CT in the assessment of abdominal volumetric FM. Men are annotated by black dots and females by white dots. In the Bland-Altman plot, the differences between methods are shown on the y-axis and mean VAT on the x-axis. Abbreviations: CT, Computed Tomography; DXA, Dual-Energy X-ray Absorptiometry; FM, fat mass.

## 4. Discussion

The main finding of this study is that DXA is a valid tool for the measurement of abdominal volumetric VAT and FM in CRC patients compared to reference method CT. The difference between methods for VAT was 76 $cm^3$ including IMAT and 201 $cm^3$ excluding IMAT, which is small in relation to the total VAT in the subjects (i.e. median $VAT_{DXA}$ is 1230 $cm^3$). The results are in line with similar validation studies in healthy subjects [2,18]. Coletta et al. conducted a validation study using cancer patients [26]. However, their results are not comparable to us as they used two different region-of-interests for $VAT_{DXA}$ and $VAT_{CT}$, in contrast to the corresponding volumes in this study. In addition to high validity, DXA also need to be precise (provide high repeatability) [27]. A precision study conducted on the same Lunar iDXA machine with CRC patients revealed high precision (%CV = 3.56) and a least significant change in VAT of 149 $cm^3$. Values below 149 $cm^3$ can be attributed to measurement variability in the DXA machine itself rather than a systematic difference between DXA and CT [44]. Similarly, a mean difference of −319 $cm^3$ in FM is a small amount relative to the total abdominal FM (i.e. median $FM_{DXA}$ is 2511 $cm^3$).

The degree of fatness may influence the validity of DXA. We observed a higher overestimation of VAT ($p = 0.015$) concurrent with an underestimation of FM ($p = 0.036$) in obese versus normal weight subjects by DXA. To demonstrate, the mean difference between methods (DXA minus CT) was 97 $cm^3$ and 337 $cm^3$ for subjects with low (500 $cm^3$) and high amounts (2000 $cm^3$) of VAT using the regression equations. A BMI of 23 and 30 $kg/m^2$ present a

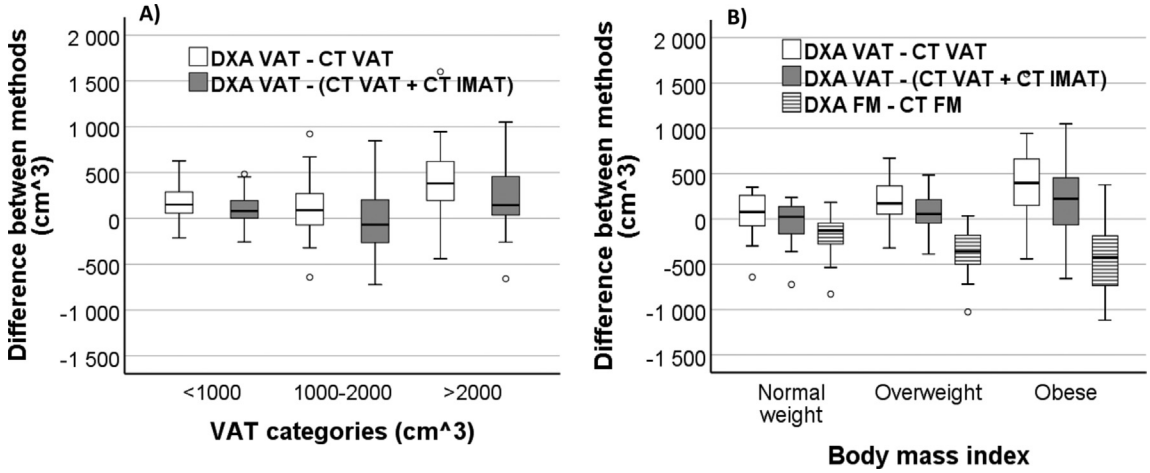


**Fig. 4.** Boxplots representing the differences between DXA and CT across groups. A) The difference between methods for the assessment of abdominal volumetric VAT (y-axis) across mean VAT categories (DXA VAT + CT VAT (+CT IMAT)/2 on x-axis). B) The difference between methods for the assessment of abdominal volumetric VAT and FM (on y-axis) stratified by WHO BMI categories (accordingly normal weight: 18.5–24.9 kg/m$^2$, overweight: 25.0–29.9 kg/m$^2$ and obese subjects: 30.0–34.9 kg/m$^2$) on the x-axis. Abbreviations: Dual-Energy X-ray Absorptiometry; VAT, visceral adipose tissue; CT, Computed Tomography; DXA; IMAT, inter- and intramuscular adipose tissue, FM, fat mass.

**Table 2**
Abdominal volumetric visceral adipose tissue versus single slice samples of body compartments with coefficients of determination ($R^2$).

| Abdominal volumes vs. single slices | Level of CT single slice and slice combinations | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | L2 | L3 | L4 | L2+L3 | L2+L4 | L3+L4 | L2+L3+L4 | p-value L3 vs. L2+L3+L4 |
| n | 65 | 66 | 46 | 65 | 45 | 46 | 45 | |
| $VAT_{CT}$ volume (cm$^3$) vs. $VAT_{CT}$ single slice | 0.94 | 0.97 | 0.94 | 0.97 | 0.97 | 0.97 | 0.98 | 0.006* |
| $VAT_{DXA}$ volume (cm$^3$) vs. $VAT_{CT}$ single slice | 0.82 | 0.87 | 0.86 | 0.86 | 0.89 | 0.90 | 0.90 | 0.004* |
| $VAT_{DXA}$ volume (cm$^3$) vs. $VAT_{CT}$ + $IMAT_{CT}$ single slice | 0.86 | 0.87 | 0.87 | 0.88 | 0.91 | 0.91 | 0.92 | <0.001* |

Abbreviations CT, Computed Tomography; VAT, visceral adipose tissue; DXA, Dual-Energy X-Ray Absorptiometry; IMAT, inter- and intramuscular adipose tissue. *p-value calculated by pairwise comparisons proposed by Meng, Rosenthal and Rubin.

difference between methods of 68 cm$^3$ and 291 cm$^3$ for VAT and −254 and −387 cm$^3$ for FM, respectively. Caution should therefore be taken when comparing subjects with very high and low amounts of VAT measured by DXA, as well as subjects in underweight or normal weight versus obese BMI groups. However, patients are rarely subject to extreme changes in BMI, even in the event of disease related malnutrition with 10% weight loss [28]. DXA can therefore safely be used in the follow-up of most patients in clinical practice.

Anthropometric measurements are not sensitive for determining the distribution of adipose tissues and muscle mass [7]. In this study, the anthropometric measurements described approximately half of the variation (31–65%) of volumetric VAT measured by DXA and CT. A study using similar instructions for WC and hip circumference measurement as us in obese and overweight patients showed that the mean difference in measurements between observers was not clinically significant [29]. We do not expect precision errors to influence our results. The risk of several cancers, cardiovascular diseases and other lifestyle-related conditions are generally found to be more associated with VAT than overall FM [30,31]. Therefore, we suggest looking beyond the commonly used anthropometric measurements like BMI. For instance, CT is integrated into cancer care for CRC patients. By using CT scans from electronic journals we can retrieve information about body composition without providing extra radiation exposure [15]. This is a unique opportunity to introduce more personalized treatment and follow-up on body composition in cancer care. In other patient groups without clinical CT data, we propose greater use of DXA. DXA is often available at hospitals as it is the main modality for diagnosis and management of osteoporosis [32], but is rarely in use for VAT or FM assessment. With DXA one can perform a whole-body scan, while whole-body estimates derived from single CT-slices usually require the use of prediction equations [33] which introduce additional uncertainty. Therefore, whole body measurement by DXA might be superior to estimates derived from a single CT slice, despite CT being described as more accurate than DXA [15]. In a clinical practice, both DXA and CT may contribute to the evaluation of nutritional status, tailoring of nutritional support or intervention [32,34] optimizing treatment [13,35] and may influence survival [4,5].

In this study we compared DXA and CT's quantification of VAT, with and without the inclusion of IMAT. There are several reasons why IMAT might partly be included in DXA's estimate of VAT. First, DXA can differentiate FM and lean fractions in areas where bone is not present [16,36]. In a whole-body scan, bones are present in 30–45% of the pixels [37,38]. In the abdominal volumes applied in the current study even less bone is present, as only the spine is displayed. Secondly, DXA does not segment or classify IMAT separately, but it is most likely included in other DXA measures of adipose tissue. The DXA software identify VAT by subtracting SAT from total FM in the android region [17,18,39]. We propose that DXA detect IMAT at some extent and embed this

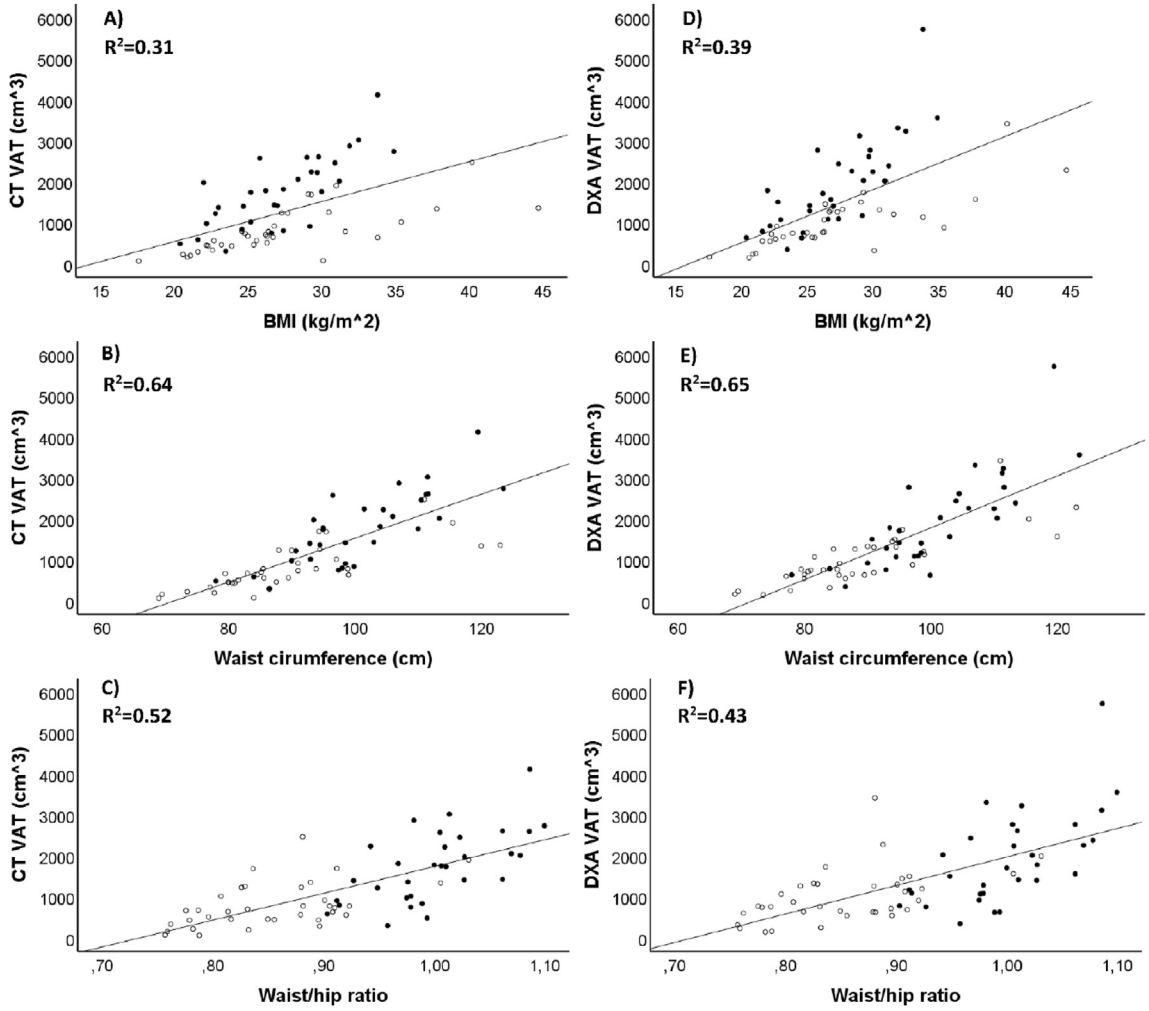


**Fig. 5.** Coefficient of determination ($R^2$) between the BMI, waist circumference and waist/hip ratio and abdominal volumetric $VAT_{CT}$ (A–C) and $VAT_{DXA}$ (D–F). Men are annotated by black dots and females by white dots. Abbreviations: CT Computed Tomography; VAT, visceral adipose tissue; DXA, Dual-Energy X-Ray Absorptiometry.

adipose tissue within VAT. This is supported by the increased explained variance (from 89% to 91%) and less mean difference between the methods (from 201 $cm^3$ to 76 $cm^3$, $p < 0.001$) when adding $IMAT_{CT}$ to $VAT_{CT}$ in the comparison against $VAT_{DXA}$. Thus, we assume that it is likely that DXA detects parts of IMAT, especially in individuals in which fat has replaced large amounts of muscle. However, more research such as cadaver and phantom studies is needed to confirm this hypothesis, and quantify the exact amounts of IMAT that is detected by DXA.

VAT measurements obtained by CT have typically been derived from a single slice. This is due to time-consuming segmentation in lack of a quantifying software [33,40]. We showed that a single slice at the L3-level was superior across the lumbar slices (explained 87–97% of volumes), similar to results from other studies [41,42]. However, the combination of three slices at L2, L3, and L4-level demonstrated significantly better explained variance (explained 90–98% of volumes) than a single L3 slice alone ($p < 0.006$ for all comparisons). Accordingly, future body composition measurement by CT should aim against using multiple slices or volumetric approaches. This requires automatic software, which will be one of our next objectives. Until then, we suggest the combination of three slices at different lumbar localizations (L2, L3, L4). This will manage the shortcomings of the single slice approach while reducing the labour-intensive work of acquiring volumetric CT data in a non-automatic manner.

The time-frame between the scans is a limitation of this study, and we cannot rule out changes in the patient's body composition in the days between the DXA and CT scan. However, additional analysis showed that time did not statistically impact the difference between methods in a univariate regression model (data not shown). Only five out of 66 patients received chemotherapy and most of the patients reported no weight change. Also, our population with TNM I-III can be considered quite stabile compared to patients with more advanced CRC. The CT segmentation was performed by a single operator and followed the established Alberta protocol. A previous study demonstrated low level of operator dependability for CT body composition segmentation which suggest that our results are reproducible [43]. Strengths of this study includes the archived power of 80%, according to our sample size estimation. As far as we know, this is the first study to investigate corresponding volumes of VAT and abdominal FM from DXA and CT in a pure CRC population.

We conclude that DXA and the combined use of three lumbar CT slices (L2-L4) are valid tools for the estimation of abdominal VAT and FM in CRC patients, when using volumetric CT as reference method. We suggest further exploration of the increased use of

DXA and CT in standard cancer care as this may add great clinical value.

## Statement of authorship (CRediT)

Dena Helene Alavi: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Roles/Writing - original draft, Writing - review & editing, Hege Berg Henriksen: Conceptualization, Data curation, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing - review & editing. Peter Mæhre Lauritzen: Conceptualization, Data curation, Investigation, Methodology, Project administration; Resources, Supervision, Validation, Writing - review & editing. Ane Sørlie Kværner: Conceptualization, Data curation, Investigation, Supervision, Writing - review & editing. Tomas Sakinis: Data curation, Resources, Software. Torgrim Mikal Langleite: Project administration, Data curation, Resources, Writing - review & editing. Christine Henriksen: Conceptualization, Writing - review & editing. Siv Kjølsrud Bøhn: Conceptualization, Project administration (of the CRC-NORDIET study), Writing - review & editing. Ingvild Paur: Conceptualization, Project administration (of the CRC-NORDIET study), Writing - review & editing. Gro Wiedswang: Conceptualization, Resources, Writing - review & editing. Sigbjørn Smeland: Conceptualization, Resources, Writing - review & editing. Rune Blomhoff: Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Writing - review & editing.

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## Declaration of competing interest

RB is a shareholder of AS Vitas. All the other authors declares no conflict of interests.

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